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(54) **ELECTRONIC TIMEPIECE AND ELECTRONIC TIMEPIECE MAIN BODY**

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(57) **ABSTRACT**

Provided is an electronic timepiece including a timepiece main body; and an exterior member attachable to the timepiece main body. The timepiece main body includes: a plurality of protrusions and a first protruding part. The plurality of protrusions which protrude outward on an outer periphery of the timepiece main body and whose apexes on extending sides form an outermost shape of the timepiece main body. The first protruding part that is formed to connect adjacent ones of the plurality of protrusions and do not protrude outward more than apexes on extending sides of the connected ones of the plurality of protrusions. The exterior member has an engaging part that is engaged with the first protruding part when the exterior member is attached to the timepiece main body.

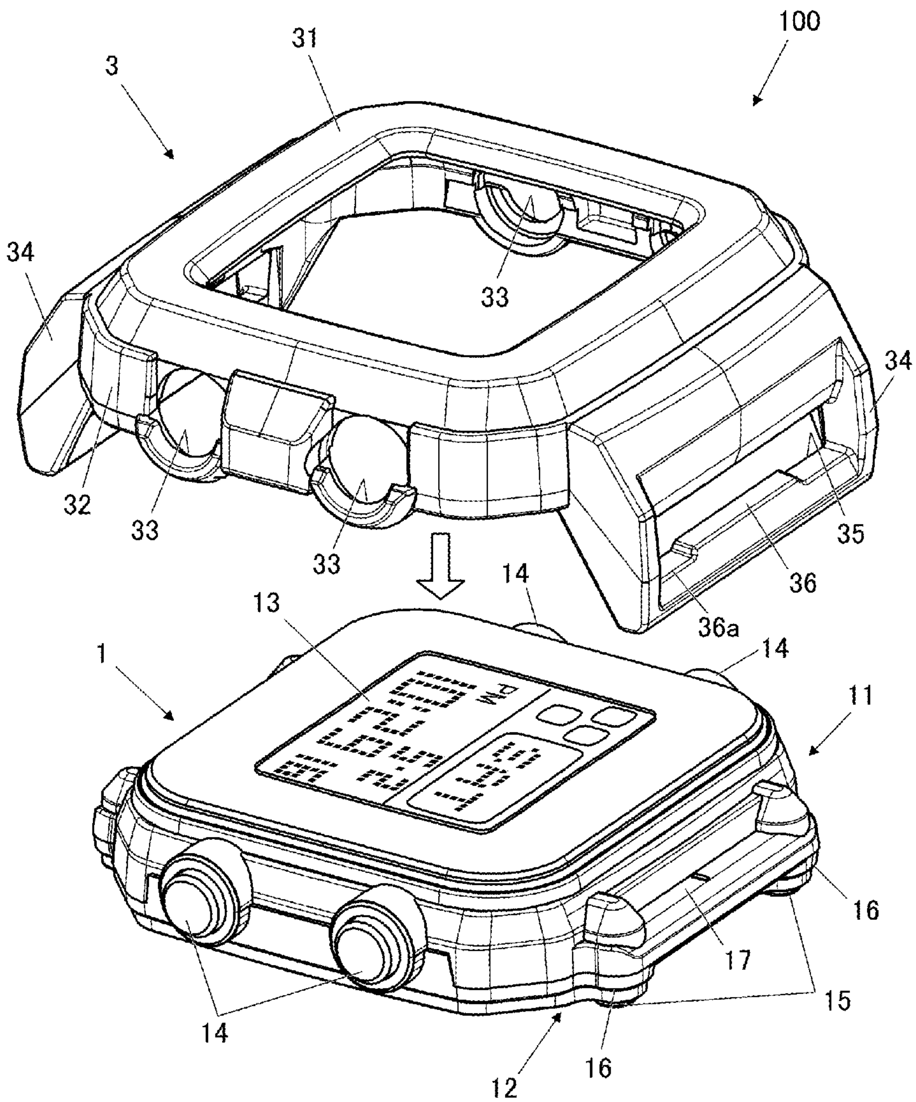


FIG.1

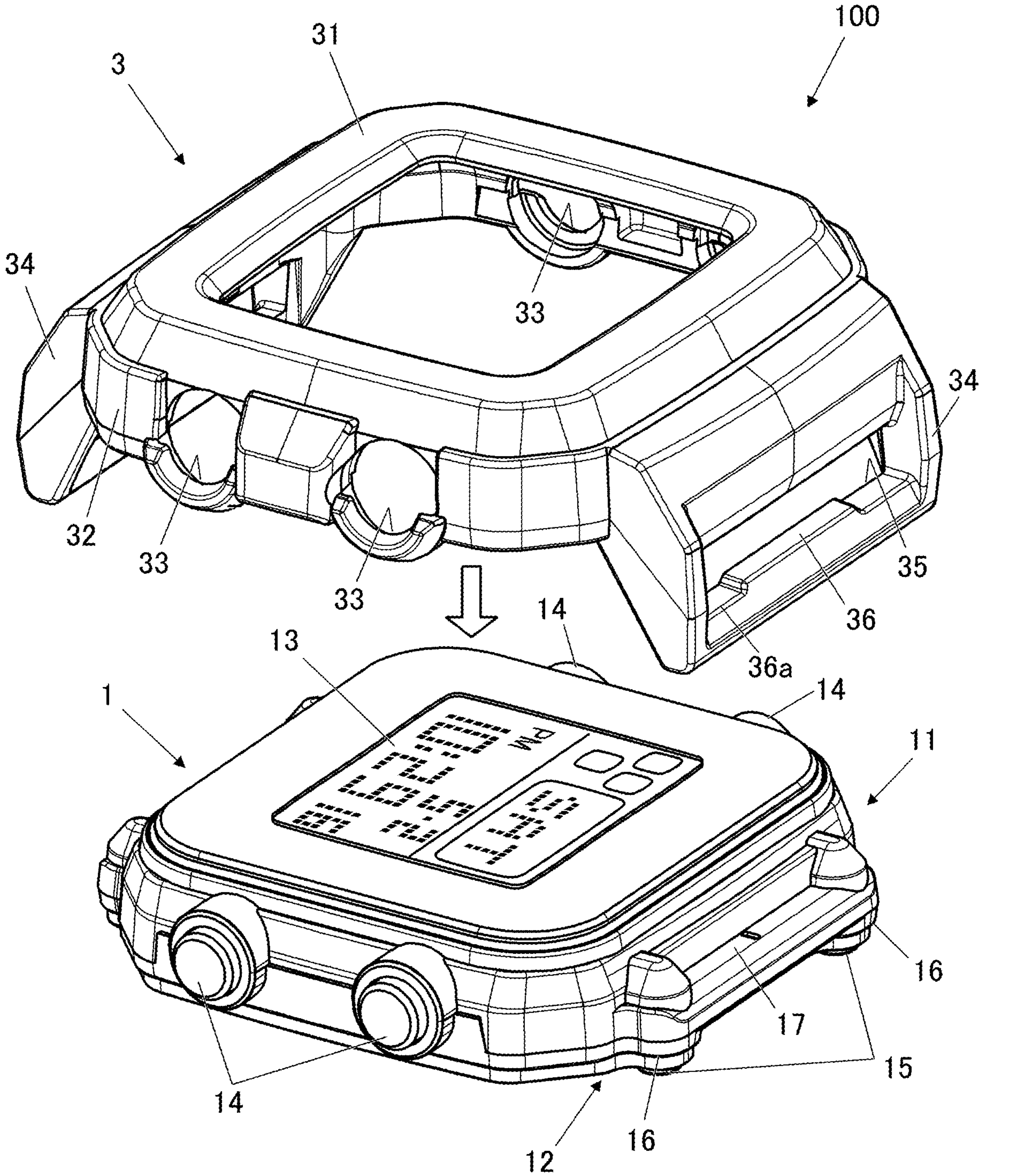
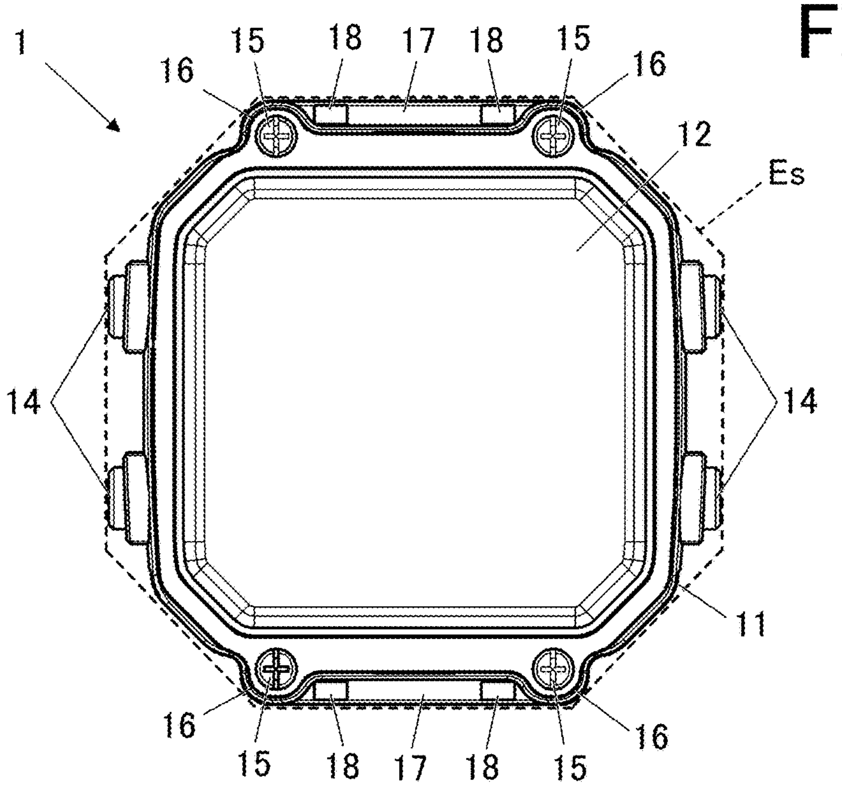


FIG.2


1
15
18
17
18
15
16
16
12
Es
14
14
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15
16

FIG.3

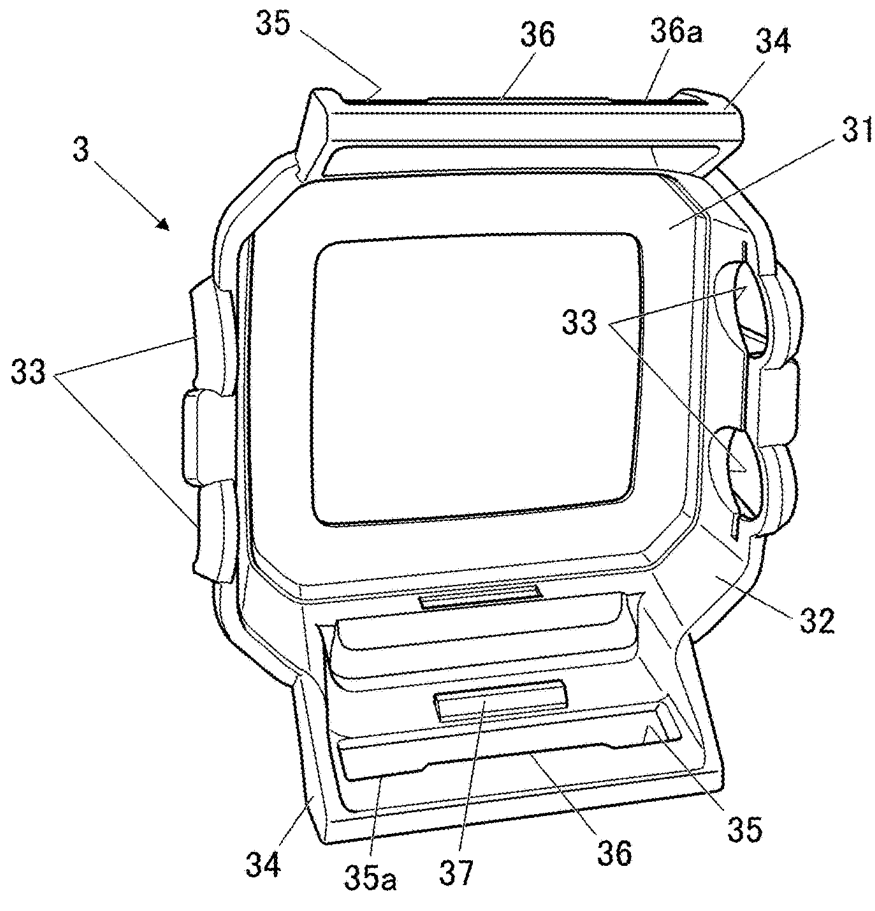

35
36
36a
34
31
3
33
33
32
34
35a
37
36
35

FIG.4

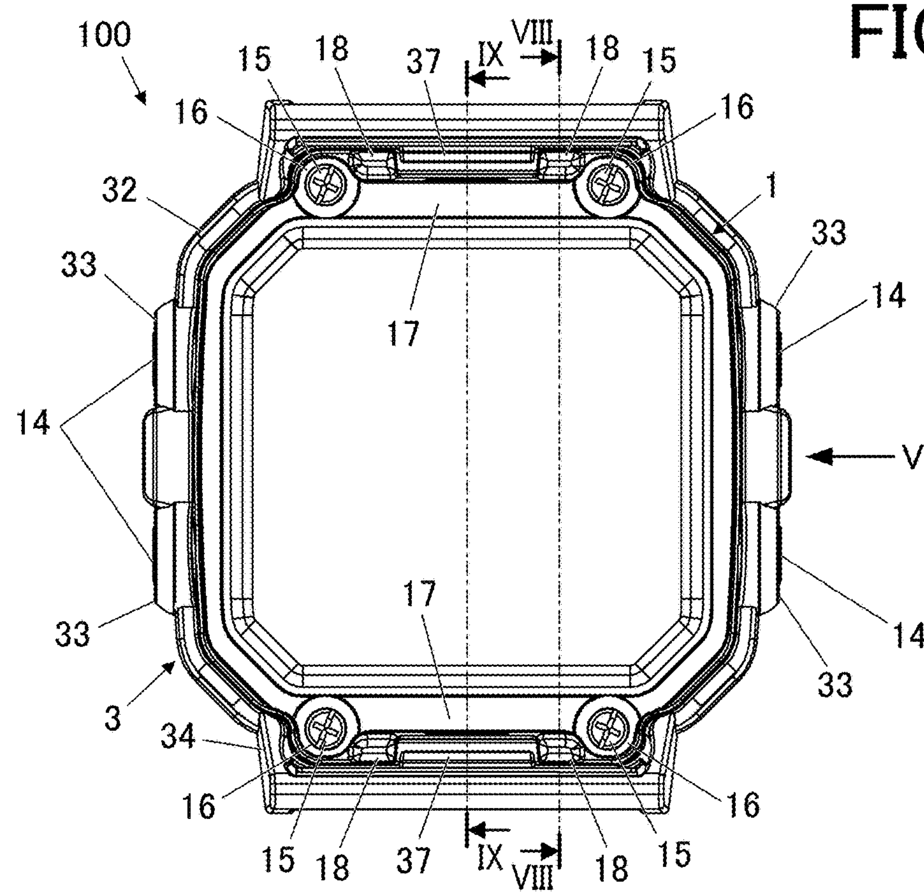

100
15
18
37
IX
VIII
18
15
16
16
32
1
33
33
14
14
17
V
17
33
14
33
3
34
16
16
15
18
37
IX
VIII
18
15

FIG.5

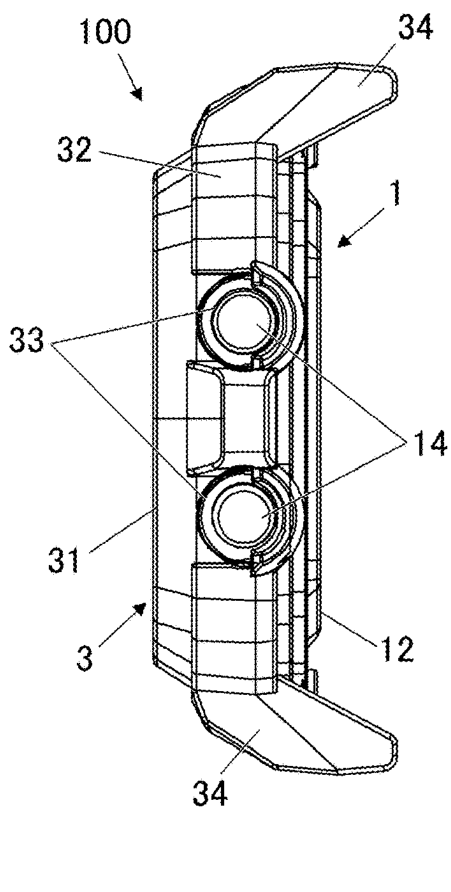

100
34
32
1
33
14
31
3
12
34

FIG.6A

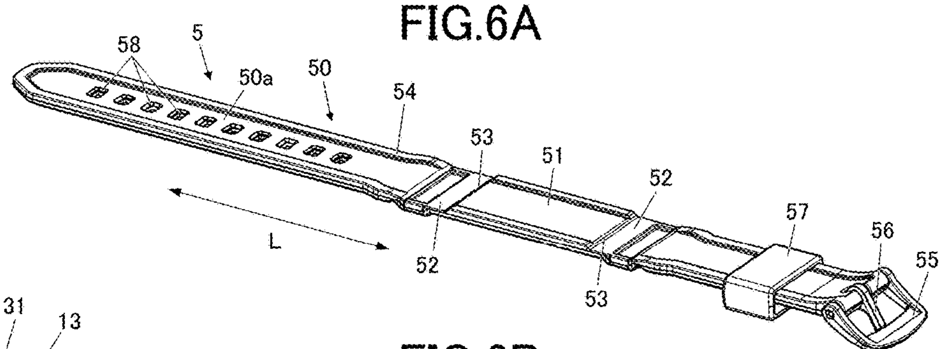
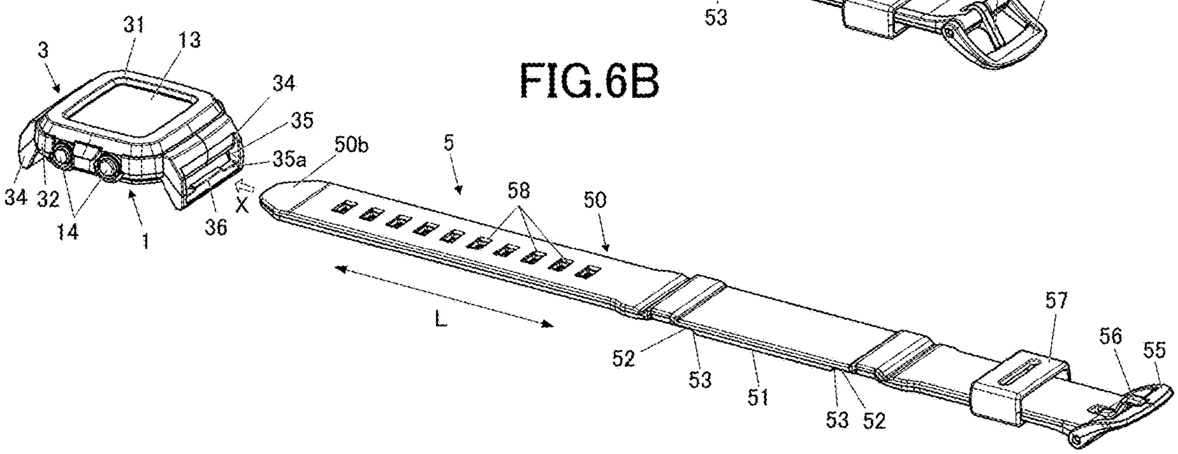

58
5
50a
50
54
53
51
52
L
52
57
56
55
53

FIG.6B


3
31
13
34
35
35a
50b
5
34
32
14
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56
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52
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51
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52

FIG.7A

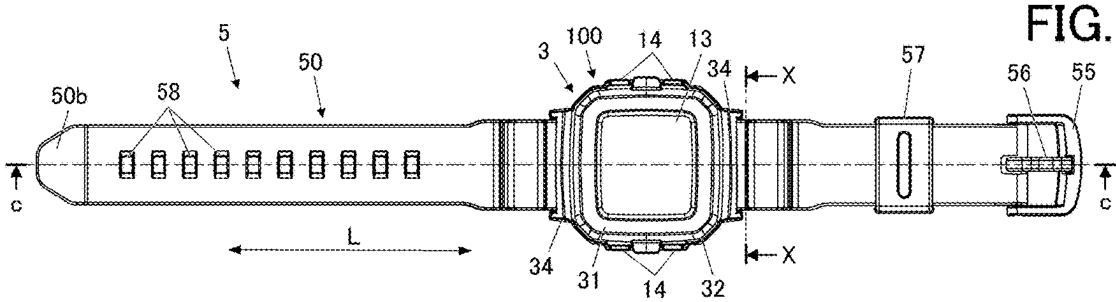


FIG.7B

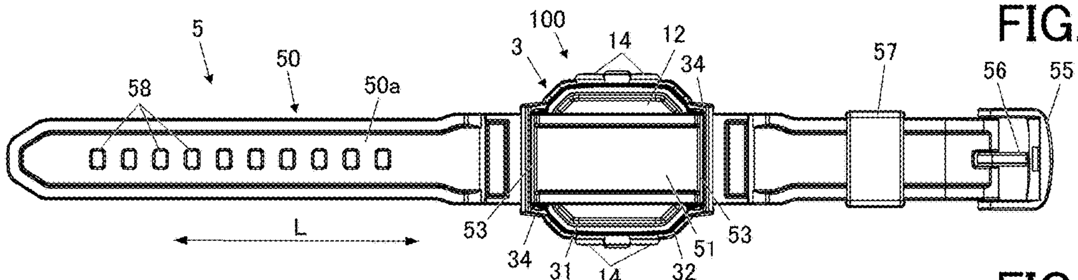
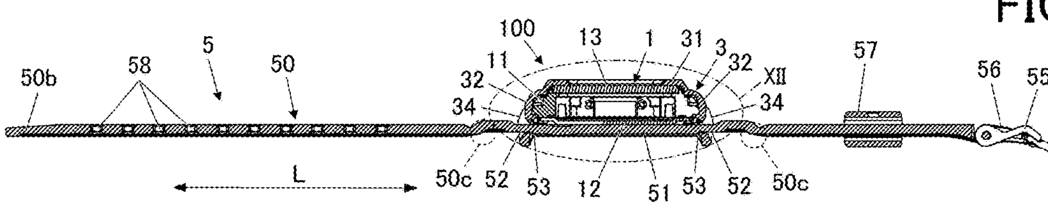


FIG.7C



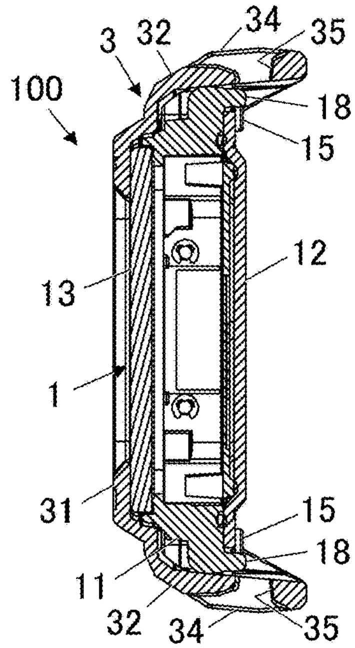


FIG.8

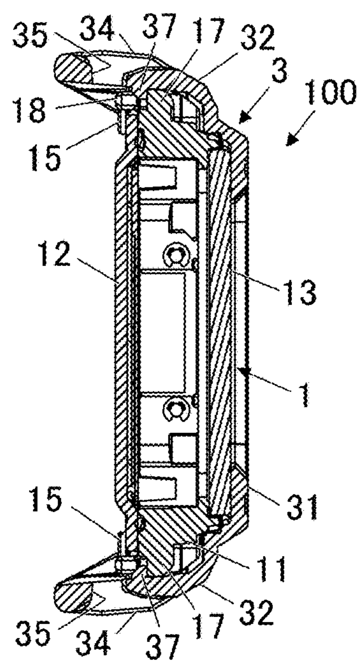


FIG.9

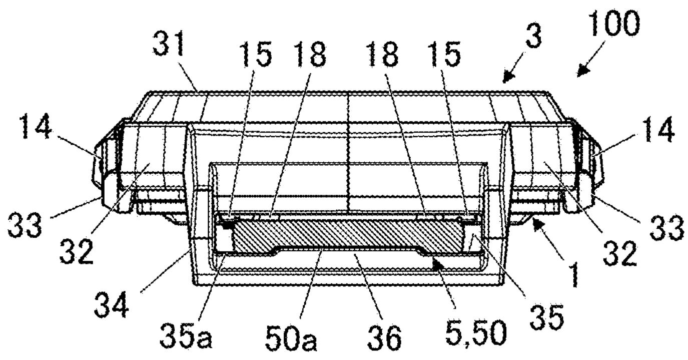


FIG.10

FIG.11

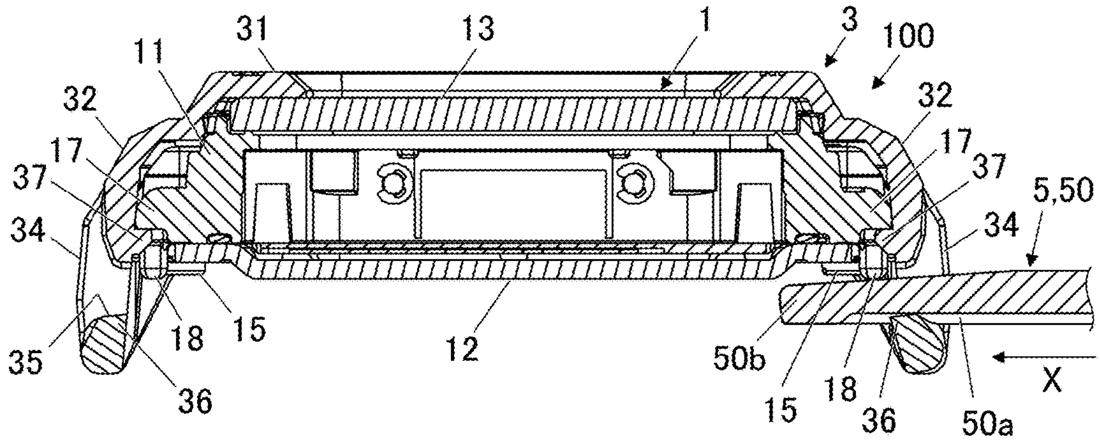


FIG.12

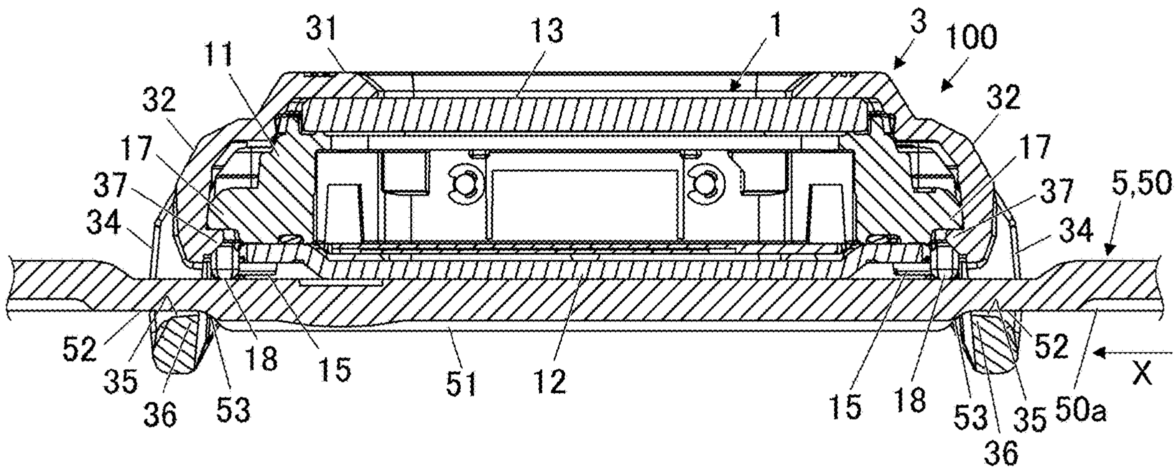


FIG.13

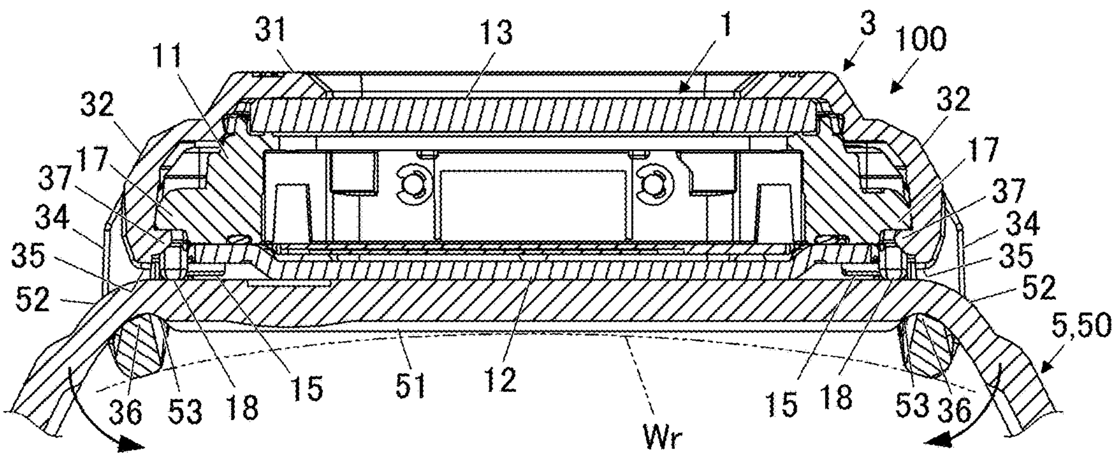


FIG.14

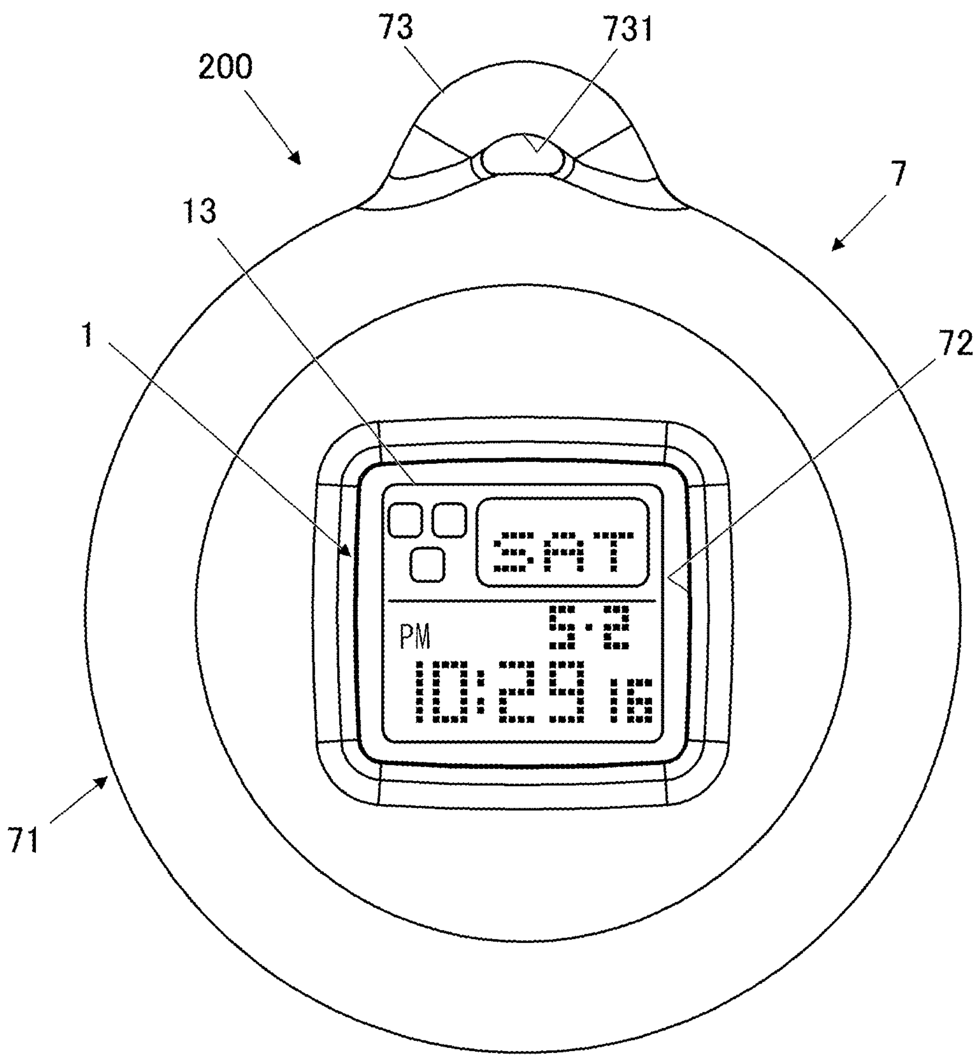
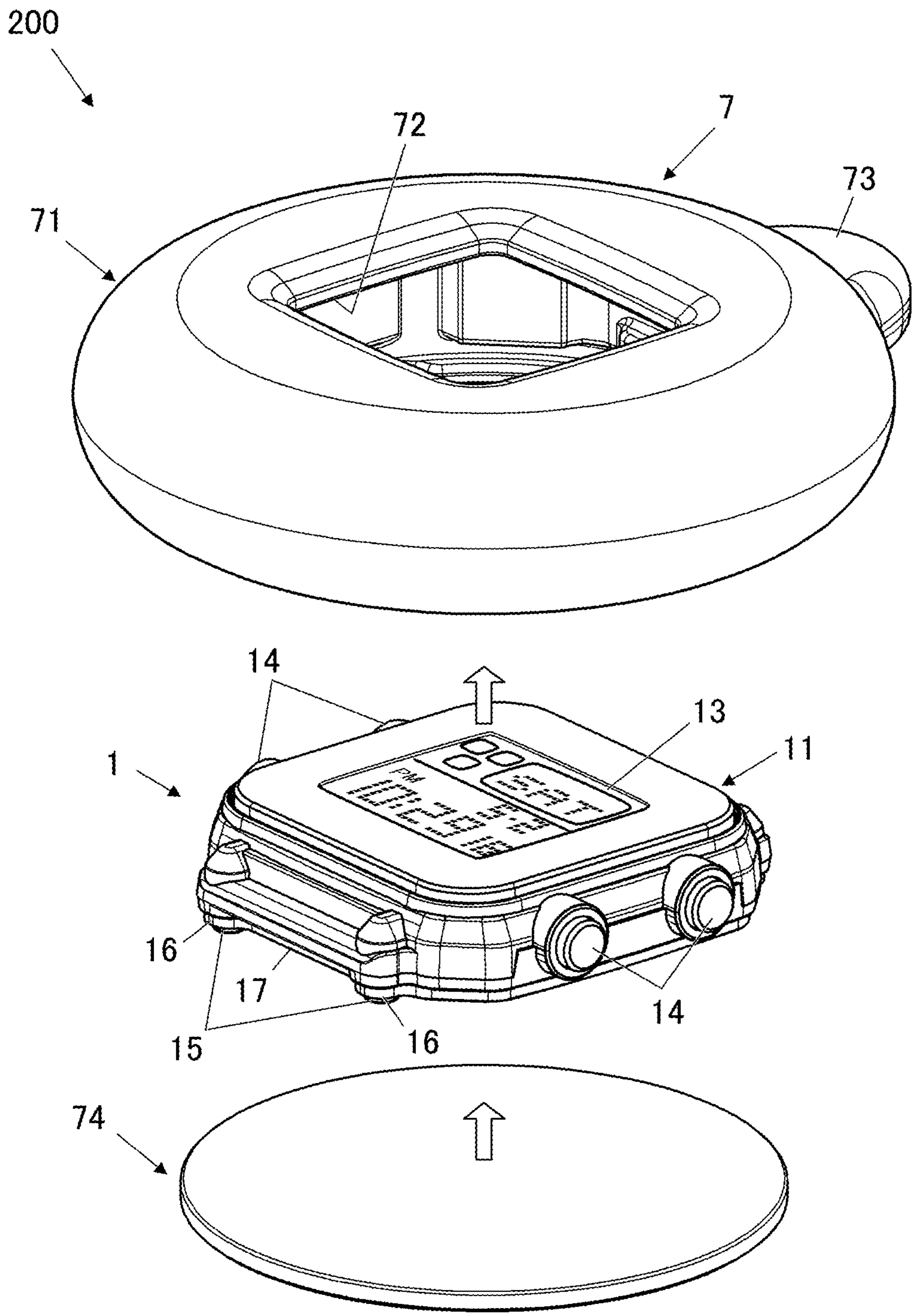


FIG.15



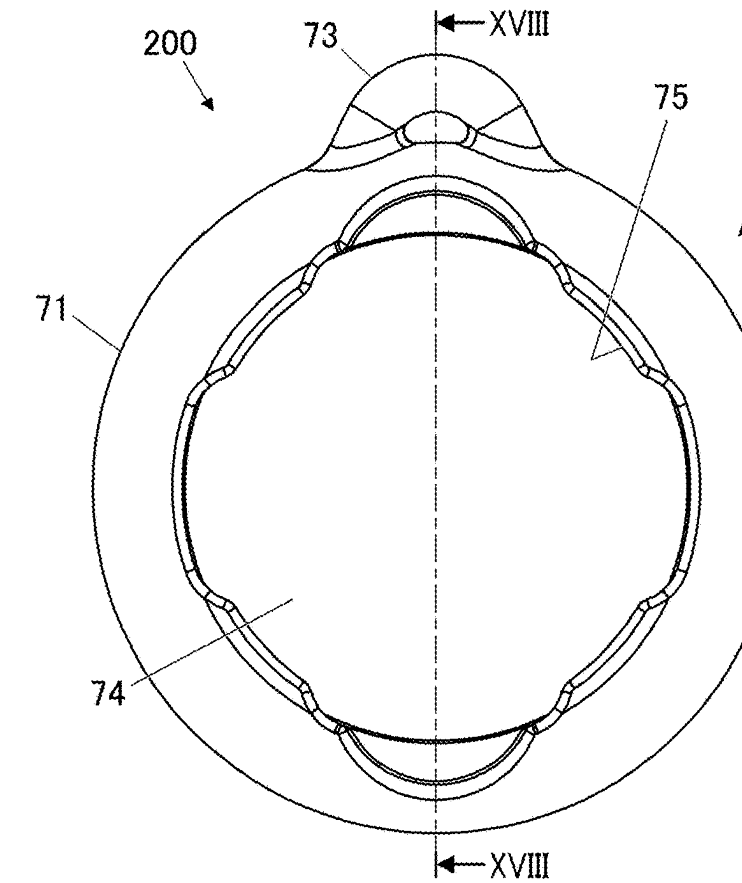
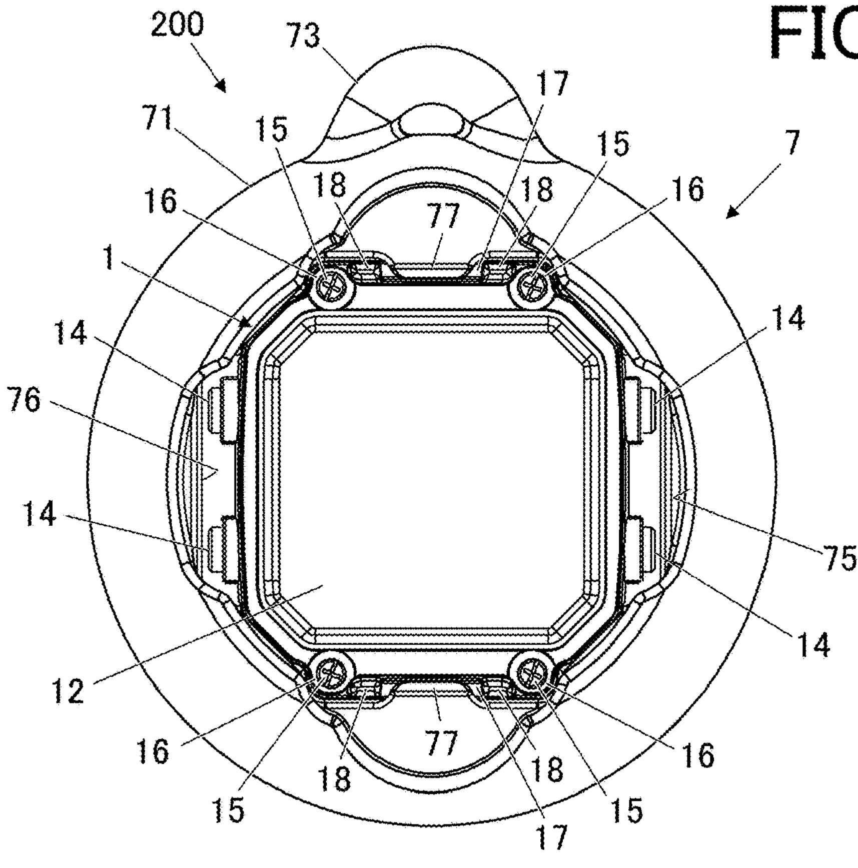

200
73
XVIII
75
7
71
74
XVIII

FIG.16


200
73
17
71
15
15
18
18
7
16
77
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75
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12
16
77
16
18
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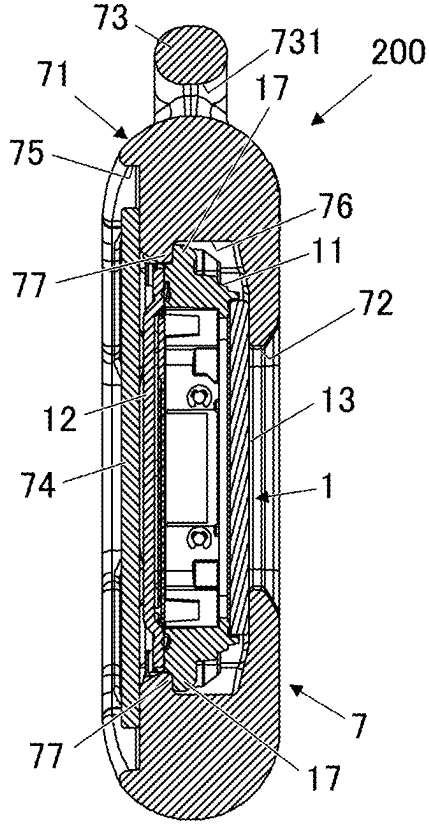
FIG.17

FIG.18



ELECTRONIC TIMEPIECE AND ELECTRONIC TIMEPIECE MAIN BODY

REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-032604, filed on Mar. 5, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present embodiment relates to an electronic timepiece and an electronic timepiece main body.

BACKGROUND ART

[0003] Conventionally, there has been known a structure in which an exterior member is attached to a timepiece main body. For example, in Japanese Laid Open Patent Publication No. 2015-010905, a locking protrusion that protrudes outward is provided on the peripheral surface of a case main body and an engaging recess that engages with the locking protrusion on the inner peripheral surface of the exterior member (a bezel in JP2005010905A) so that both parts are engaged with each other to attach the bezel to the case main body.

SUMMARY OF INVENTION

[0004] To solve the above-mentioned problem, the electronic timepiece according to the present disclosure includes:

[0005] a timepiece main body; and

[0006] an exterior member attachable to the timepiece main body,

[0007] wherein the timepiece main body includes:

[0008] a plurality of protrusions which protrude outward on an outer periphery of the timepiece main body and whose apexes on extending sides form an outermost shape of the timepiece main body; and

[0009] a first protruding part that is formed to connect adjacent ones of the plurality of protrusions and do not protrude outward more than apexes on extending sides of the connected ones of the plurality of protrusions,

[0010] wherein the exterior member has an engaging part that is engaged with the first protruding part when the exterior member is attached to the timepiece main body.

BRIEF DSCRIPTION OF DRAWINGS

[0011] FIG. **1** is a perspective view of a major part showing how an exterior member according to an embodiment is attached to the frontal face side of a timepiece main body according to the embodiment.

[0012] FIG. **2** is a plan view of the back face side of the timepiece main body according to the embodiment.

[0013] FIG. **3** is a perspective view of the exterior member according to the embodiment as viewed from the inside.

[0014] FIG. **4** is a plan view of the timepiece main body with the exterior member attached thereto as viewed from the back face side.

[0015] FIG. **5** is a lateral view of the timepiece main body with the exterior member attached as viewed from a direction of an arrow V in FIG. **4**.

[0016] FIG. **6**A is a perspective view of a band according to the embodiment as viewed from the back side.

[0017] FIG. **6**B is a perspective view of a major part showing how the band shown in FIG. **6**A is attached to the timepiece main body with the exterior member attached.

[0018] FIG. **7**A is a frontal view of a timepiece according to a first embodiment.

[0019] FIG. **7**B is a rear view of the timepiece shown in FIG. **7**A.

[0020] FIG. **7**C is a cross-sectional view taken along a line c-c of the timepiece shown in FIG. **7**A.

[0021] FIG. **8** is a cross-sectional view taken along a line VIII-VIII of the timepiece main body with the exterior member attached in FIG. **4**.

[0022] FIG. **9** is a cross-sectional view taken along a line IX-IX of the timepiece main body with the exterior member attached in FIG. **4**.

[0023] FIG. **10** is a cross-sectional view taken along a line X-X of the timepiece shown in FIG. **7**A.

[0024] FIG. **11** is a cross-sectional side view of the major part showing how the band is attached to the exterior member attached to the timepiece main body.

[0025] FIG. **12** is a cross-sectional side view of the major part showing a state in which the band is attached to the exterior member attached to the timepiece main body.

[0026] FIG. **13** is a cross-sectional side view of the major part showing an illustration of how the band of the timepiece in FIG. **12** is put around a wrist of a wearer.

[0027] FIG. **14** is a frontal view of a timepiece according to a second embodiment.

[0028] FIG. **15** is a perspective view showing how the timepiece according to the second embodiment is assembled by attaching an exterior case to the timepiece main body.

[0029] FIG. **16** is a plan view of the back face side of the timepiece shown in FIG. **14**.

[0030] FIG. **17** is a plan view showing a state in which a back cover is detached from the timepiece shown in FIG. **16**.

[0031] FIG. **18** is a cross-sectional view taken along a line XVIII-XVIII of the timepiece shown in FIG. **16**.

DETAILED DESCRIPTION

[0032] Embodiments of an electronic timepiece and an electronic timepiece main body according to the present disclosure are described with reference to the drawings. Various limitations technically preferable for carrying out the present disclosure are provided in the following embodiments. However, the scope of the present invention is not limited to the embodiments below or illustrated examples. In the embodiments hereinbelow, the electronic timepiece is simply referred to as a "timepiece."

[0033] As shown in FIGS. **1**, **14**, etc., the timepiece according to the present embodiment includes an electronic timepiece main body (hereinafter simply referred to as a "timepiece main body **1**"), an exterior member attachable to the timepiece main body **1**. In the embodiment, the exterior member is a first member or a second member selectively attached to the timepiece main body **1**. The first member is a bezel **3** to which a band **5** can be attached, and the second member is an exterior case **7**. The bezel **3** and the exterior case **7** can be attachable to and detachable from the timepiece main body **1**. In the following description, a "timepiece **100**" in the first embodiment is in a state where the bezel **3** is attached to the timepiece main body **1**, and a "timepiece **200**" in the second embodiment is in a state where the exterior case **7** is attached to the timepiece main body **1**. The configuration of the timepiece main body **1** is common in the first and second embodiments. Thus, the timepiece main body **1** is first described as a configuration common in the first and second embodiments.

[0034] The timepiece main body **1** according to the embodiment includes a main body case **11** which houses a timepiece module (not shown in the drawings) inside and a back cover **12** attached to the back side of the main body case **11**. The main body case **11** is formed of a kind of synthetic resin, for example, and the back cover **12** is formed of a kind of metal such as a stainless steel (SUS), for example.

[0035] The timepiece main body **1** in the embodiment is formed to have an external shape of a substantially rectangle in a plan view, as shown in FIG. **2**. The timepiece main body **1** has outwardly extending protrusions on the outer side. The apexes of the protrusions on the extending side form an outermost shape Es of the timepiece main body **1**. In FIG. **2**, the outermost shape Es of the timepiece main body **1** is shown by a dashed line. As shown in FIG. **2**, the timepiece main body **1** according the embodiment is approximately symmetrical vertically and laterally, and the forms and the lengths of the sides facing each other (corresponding to the 6 and 12 o'clock sides and the 3 and 9 o'clock sides) are almost the same. (Hereinafter, when referred to a "o'clock" side, it indicates each side of the timepiece as on a dial of an analog watch.) The timepiece main body **1** is in the same shape when rotated 180 degrees.

[0036] The timepiece module housed in the main body case **11** includes components necessary for the watch functions such as a display unit that displays information such as time, a driver that drives the display, and a circuit that controls these (none of which are shown in the drawings), for example. In the present embodiment, the display includes a display panel **13** (see FIG. **1**) exposed to the front face side of the timepiece main body **1** and displays various pieces of information such as time, date, and day on the display panel **13**. The display panel **13** is constituted of, for example, a liquid crystal display (LCD), an organic electroluminescence display, or another flat display. The display unit is not limited to the one with the display panel **13** displaying the date and the like. It may be a display unit of an analog type with hands and a dial or a so-called combination model with a display unit capable of analog and digital display.

[0037] Operation units **14** are provided on the lateral sides of the main body case **11**. The operation units **14** are parts through which the user inputs various operations, which are a push button, a crown, and the like, for example. In FIGS. **1** and **2**, two of the operation units **14** are provided on the 3 o'clock side and other two are provided on the 9 o'clock side, for example. The operation units **14** are second protruding parts on the timepiece main body **1**. The operation units **14** extend outward on the 3 and 9 o'clock sides of the timepiece main body **1**, and as shown by a broken line in FIG. **2**, the apexes of the protrusions on the extending sides form the outermost shape Es on the 3 and 9 o'clock sides of the timepiece main body **1**.

[0038] As shown in FIG. **2**, the back cover **12** is attached to the main body case **11** by being screwed on the opposing two sides of the timepiece main body **1**. Screw hole spaces **16** are provided at the portions where the main body case **11** and the back cover **12** are screwed together. In the embodiment, the back cover **12** is screwed with two screws **15** each on the 6 and 12 o'clock sides. That is, the portions where the back cover **12** is screwed (the screw hole spaces **16**) are on the sides different from those on which the operation units **14** are provided.

[0039] The back cover **12** is screwed not to interfere with the timepiece module housed in the main body case **11**. Thus, the screw hole spaces **16** are protrusions provided outside the space where the timepiece module is housed and formed on the outer periphery of the timepiece main body **1**. The screw hole spaces **16** as the protrusions extend outward from the timepiece main body **1** on the 6 and 12 o'clock sides, and as shown by a broken line in FIG. **2**,, the apexes of the protrusions on the extending side form the outermost shape Es on the 6 and 12 o'clock sides of the timepiece main body **1**.

[0040] First protruding parts **17** are formed between adjacent ones of the screw hole spaces **16** as the protrusions by adding a material to bridge the gap between the adjacent screw hole spaces **16**. The first protruding parts **17** are provided in a range that does not go over the line connecting the apexes on the protruding side of the corresponding screw hole spaces **16** (i.e., inside the outermost shape Es of the timepiece main body **1**). In the embodiment, the first protruding part **17** is formed to connect the adjacent ones of the screw hole spaces **16** (protrusions) on the same side so that the first protruding part **17** and the screw hole spaces **16** are aligned substantially flush as shown in FIG. **2**, for example. Thus, the outermost shape Es of the timepiece main body **1** can be a shape of an excellent design with less unevenness. The shape of the first protruding parts **17** is not limited to the illustrated examples, and may be recessed inward from the line connecting the adjacent ones of the screw hole spaces **16** (protrusions).

[0041] For the timepiece **100** according to the embodiment, the band **5** is positioned to go through the back face side of the timepiece main body **1** (the side on which the back cover **12** is provided) when the band **5** is attached thereto. The screws **15** that fix the back cover **12** are positioned on a path where the band **5** passes when the band **5** is being attached. The screws **15** are provided on the back face side of the timepiece main body **1** and formed of a metal such as stainless steel (SUS) or the like. Thus, in a case where the band **5** is in contact with the screws **15** (especially screw head) the front surface of the band **5** may be rubbed or scraped and damaged.

[0042] In this regard, protective protrusions **18** which protrude from the line connecting the apexes (screw heads) of the screws **15** are provided in the vicinity of the screws **15**, on the back side of the first protruding part **17** (the back face side of the timepiece main body **1**) on the timepiece main body **1** according to the embodiment (see FIGS. **2**, **4**, and **10**). How much the protective protrusions **18** protrude from the line connecting the apexes of the screws **15** (screw heads) is determined as appropriate. If the protective protrusions **18** protrude too much, the protective protrusions **18** obstruct the passage of the band **5** when the band **5** is being attached. Thus, as shown in FIG. **8**, for example, it is preferable that the protective protrusions **18** go over the line connecting the apexes of the screws **15**, but do not go below the surface on the back side of the back cover **12**.

[0043] The protective protrusions **18** are illustrated as a quadrangular prism having a substantially rectangular shape in plan view in FIG. **2**, etc., but the shape or the like of the protective protrusions **18** is not limited to the illustrated example. For example, the protective protrusions **18** may be in a cylindrical shape with a circular or elliptical shape in plan view, or the like. The protective protrusions **18** have chamfered or round edges at corners on the surface on the side opposing the band **5** so that the surface of the band **5** is not easily caught.

FIRST EMBODIMENT

[0044] The exterior member (the first member) constituting the timepiece main body **100** according to the first embodiment is the bezel **3** attached to the timepiece main body **1**. The bezel **3** has a first opening formed at one end at least. The band **5** is inserted through the first opening, and a first protruding part **36** that protrudes inside the first opening is provided on an edge **35***a* that faces the back side of the band **5** (the surface not visible when the band is worn on the arm, the lower side in FIG. **10**) when the band **5** is inserted through the first opening. In the present embodiment, the bezel **3** is attached to the timepiece main body **1** so as to surround the outer periphery of the timepiece main body **1**. The bezel **3** according to the present embodiment is described in detail below with reference to FIGS. **1**, **3**, etc.

[0045] As shown in FIGS. **1** and **3**, the bezel **3** according to the present embodiment includes a frame-shaped part **31**, a lateral part **32**, and extending parts **34**. The frame-shaped part **31** is disposed on the front face side of the timepiece main body **1** when the timepiece main body **1** is worn. The lateral part **32** goes downward from the frame-shaped part **31** and is provided on the lateral part of the timepiece main body **1** when the timepiece main body **1** is worn. The extending parts **34** extend downward from the back face (the back cover **12** in the present embodiment) of the timepiece main body **1** when the bezel **3** is attached to the timepiece main body **1**. In the present embodiment, as shown in FIGS. **4**, **5**, etc., a pair of extending parts **34** is provided so as to face each other with the timepiece main body **1** in between when the bezel **3** is attached to the timepiece main body **1**.

[0046] In the embodiment, as shown in FIG. **1**, the bezel **3** is attached to the timepiece main body **1** from the front face side, and the extending part **34** is provided on the side of the timepiece main body **1** on which the first protruding part **17** is provided. That is, in the embodiment, the extending part **34** is provided each on the 6 and 12 o'clock sides when the bezel **3** is attached to the timepiece main body **1**. The bezel **3** is formed of a urethane resin such as polyurethane (PUR), for example. The bezel **3** is preferably formed of a synthetic resin such as urethane resin and has flexibility overall. At least the extending part **34** can be elastically deformed. But the material for forming the bezel **3** is not limited to this.

[0047] When the bezel **3** is attached to the timepiece main body **1**, at least part of the display panel **13** of the timepiece main body **1** is exposed in the frame-shaped part **31** so that various information such as the time is visible on the display. In the lateral part **32**, engagement holes **33** are formed respectively at positions corresponding to the operation units **14** of the timepiece main body **1** when the bezel **3** is attached to the timepiece main body **1**. In the embodiment, the engagement holes **33** are through holes which the operation units **14**, as a second protruding part, are inserted through, and when the bezel **3** is attached to the timepiece main body **1**, the operation units **14** are fitted into the engagement holes **33**. In the state where the bezel **3** is attached to the timepiece main body **1**, the operation units **14** are exposed from the engagement holes **33** and the operation units **14** can be externally operated while the bezel **3** is attached.

[0048] The pair of the extending parts **34** respectively has a pair of openings **35**. One of the openings **35** provided on one of the extending parts **34** is referred to as the first opening and the other one of the openings **35** provided on the other one of the extending parts **34** is referred to as a second opening. The band **5** is inserted through the openings **35** when the band **5** is worn, and has a width equal to or greater than the width of the band **5**. The openings **35** only need to face each other with the timepiece main body **1** in between when the bezel **3** is attached to the timepiece main body **1**, and in a case where the lateral part **32** extends downward from the back face of the timepiece main body **1**, for example, the openings may be formed at the positions where the lateral parts **32** face each other.

[0049] The protruding part **36** (the first protruding part) that protrudes inside the opening **35** (the first opening) is provided on the edge **35***a* of the opening **35** which faces the surface of the back side of the band **5** when the band **5** is inserted to the opening **35**. The first protruding part is provided on the first opening on one end of the bezel **3**, but in the present embodiment, the first opening and the second opening are provided at positions facing each other with the timepiece main bod y1 in between, and these openings **35** respectively have the protruding parts **36** (the first protruding part and the second protruding part) that protrude inside the openings **35** (the first opening and the second opening). The protruding part **36** may be provided only on one of the pair of the openings **35** (the first opening and the second opening).

[0050] The band **5** inserted through the opening **35** includes a second part **52** which is a first planar part on the back side as described later and a first part **51** which is a second planar part protruding from the level of the second part **52** (the first planar part on the back side). That is, the first part **51** as the second planar part and the second part **52** as the first planar part have different heights in the thickness direction, and the first part **51** swells toward the back side more than the second part **52** and is higher in the thickness direction. As described later, the boundary part between the first part **51** as the second planar part and the second part **52** as the first planar part forms a step part **53**. In the embodiment, the first part **51** is just thick enough to allow itself to pass through the part of the bezel **3** where the protruding part **36** is provided or thicker than that (first thickness). However, the protruding part **36** can be elastically deformed, and when the first part **51** gets caught when the band **5** is inserted to the opening **35**, the protruding part **36** can be bent along the insertion direction X (see FIG. **6**, etc.), allowing the first band **5** to pass. The protruding part **36** gets back to the original state when it is disposed at the second part **52**, and is fitted to the step part **53** which is the boundary part between the first part **51** and the second part **52**. The tip of the protruding part **36** preferably has a chamfered or round edge on the upstream side in the insertion direction X of the band **5** so that the band **5** can be easily inserted. The protruding part **36** is preferably angled (inclined) slightly toward the insertion direction X of the band **5**.

[0051] The bezel **3** is also provided with a member-side locking part stopped by the first protruding part **17** at a position corresponding to the first protruding part **17** when the bezel **3** is attached to the timepiece main body **1**. In the present embodiment, the member-side locking part is a first engaging part(s) **37**. The first engaging parts **37** are provided on the inner sides of the pair of the extending parts **34** and above the openings **35**, as shown in FIG. **3**. As shown in FIG. **9**, etc., the first engaging part **37** is a hook-shaped or hook-like claw or protrusion that protrudes toward the inside of the bezel **3**, and is positioned so as to be in contact with the first protruding part **17** from below (the underside, the back side in FIG. **1**) when the bezel **3** is attached to the timepiece main body **1** so as to surround the outer periphery of the timepiece main body **1**.

[0052] As described above, in the present embodiment, the bezel **3** as the first ember has: the first engaging parts **37** as the member-side locking part provided at the positions corresponding to the first protruding parts **17**; and the engagement holes **33** provided at the positions corresponding to the operation units **14**. In the state where the bezel **3** is attached to the timepiece main body **1**, as the first engaging parts **37** engage with the first protruding parts **17** and the operation units **14** are fitted to the engagement holes **33**, the bezel **3** is locked to the timepiece main body **1**.

[0053] Furthermore, in the first embodiment, as the band **5** is connected to the bezel **3** attached to the timepiece main body **1**, the timepiece **100** functions as a wristwatch that can be worn on the wrist Wr (the wrist Wr is shown by a dotted line in FIG. **13**) as the target wearer. As shown in FIG. **6**A, the band **5** includes a band main body **50**, and a buckle **55** and a pin **56** on one end of the band main body **50**. A free loop **57** through which the other side of the band **5** is inserted when the band **5** is worn on the target wearer (the wrist Wr) is provided on the side of the band main body **50** on which the buckle **55** etc. are provided.

[0054] The band main body **50** is formed of any of a variety of relatively flexible synthetic resins including urethane resins such as polyurethane (PUR), silicone resins, and fluorine-containing rubber, for example, and is formed in the shape of a long strip with flexibility. The band main body **50** has a sweat escape shape part **50***a* at a central part in the width direction recessed than the surrounding other part, and the band **5** is prevented from sticking to the skin when the band **5** is worn on the wrist Wr, etc.

[0055] As shown in FIGS. **6**A, **6**B, etc., the band main body **50** includes the first part **51** with the first thickness approximately in the center in the longitudinal direction L and the second parts **52** thinner than the first part **51** on the both sides of the first part **51** in the longitudinal direction L. The boundary part between the first part **51** and the second part **52** is the step part **53**. In the state where the bezel **3** is attached to the timepiece main body **1**, the band **5** is inserted through the pair of the openings **35** of the bezel **3** at the positions facing each other with the timepiece main body **1** in between from the side of an insertion-side end **50***b* of the band main body **50** (see FIGS. **6**B and **11**), and as shown in FIGS. **7**A to **7**C, the band **5** is disposed so as to go on the back side of the timepiece main body **1**.

[0056] The band main body **50** has an uneven part **50***c* where there is a step(s) on the back (for example, see the part surrounded by a dashed line in FIG. **7**C) for improving the design and fitting the wrist Wr when the band **5** is worn on the wrist Wr. However, the uneven part **50***c* like this is angled more slightly than the step part **53** with respect to the reference level (the surface of the band main body **50**), which makes it easier to pull the band **5** into the opening **35** and makes it less likely to get the band **5** stuck. Further, the band main body **50** has some parts that are thicker on the front side and some parts that are thicker on the back side, but the thickness of the band main body **50** is generally uniform overall. This makes it easier to pull the band **5** through the opening **35** of the bezel **3** to attach or detach the band **5**. As the thickness of the entire band is mostly uniform, it is easier to apply decorative surface printing or the like to the band **5**.

[0057] In the embodiment, the insertion-side end **50***b* has a tapered shape, thinner and narrower toward the ends. This makes it possible to more smoothly insert the band **5** (the band main body **50**) into the opening **35**. In FIGS. **6**B, **11**, and **12**, the direction in which the band **5** is inserted to the opening **35** of the bezel **3** is indicated as an insertion direction X. In the state where the first part **51** is on the back face side of the timepiece main body **1** (as shown in FIG. **12**), the protruding part **36** protruding inside the opening **35** of the bezel **3** engages with the step part **53** of the band **5**. The band **5** is thereby supported to the bezel **3**. As the band **5** is inserted to the back face side of the timepiece main body **1** like that, the timepiece main body **1** is sandwiched between the bezel **3** and the band **5** (as shown in FIGS. **7**B, **7**C, and **12**). Thus, the engagement state of the timepiece main body **1** and the bezel **3** is more steadily supported, making it possible to prevent the timepiece main body **1** from coming off the bezel **3**.

[0058] Next, the effects of the timepiece **100** as the electronic timepiece according to the first embodiment are described with reference to FIGS. **10** to **13**. In the first embodiment, when the bezel **3** as the first member is attached to the timepiece main body **1**, the bezel **3** covers the timepiece main body **1** from the front face side as shown in FIG. **1** so that at least part of the display panel **13** is exposed in the framed area of the frame-shaped part **31**. Further, while the bezel **3** is being expanded slightly, the operation units **14** provided on the 3 and 9 o'clock sides of the timepiece main body **1** are respectively fitted into the engagement holes **33** of the bezel **3** so that the protruding tips (the button head parts) are exposed from the engagement holes **33** to be operable. The bezel **3** is locked to the timepiece main body **1** as the first engaging parts **37** as the member-side locking part are engaged with the first protruding parts **17** provided on the 6 and 12 o'clock sides.

[0059] Once the bezel **3** is locked to the timepiece main body **1**, the band main body **50** of the band **5** is inserted to the opening **36** of the bezel **3** from the side of the timepiece **100** on which the buckle **55** and the pin **56** of the band **5** are provided (the 12 o'clock side in the embodiment), as shown in FIG. **6**B. As shown in FIG. **11**, the extending parts **34** (the protruding part **36** provided on the extending part **34**) are elastically deformable, and when the band **5** is inserted to the opening **35**, the extending part **34** is appropriately inclined along the insertion direction X to pass the band **5**. When the band **5** is pulled through until the first part **51** corresponds to the back face side of the timepiece main body **1** and the protruding part **36** is positioned at the second part **52** thinner than the first part **51**, the protruding part **36** is back to the original state and engages with the step part **53** as shown in FIG. **12**.

[0060] In the present embodiment, as shown in FIG. **10**, the protruding part **36** is fitted to the sweat escape shape part **50***a* recessed relative to the surrounding part in the cross section of the band main body **50** as shown in FIG. **10**. Therefore, the sweat escape shape part **50***a* acts as a guide, making it possible to pull the band main body **50** along the insertion direction X. The band **5** is pulled into the opening **35** with the front side of the band main body **50** facing the back face side of the timepiece main body **1**. The protective protrusion **18** in the vicinity of the screws **15** exposed to the back face side of the timepiece main body **1**. This prevents the surface of the band main body **50** from interfering with the screw **15** and being damaged.

[0061] As shown in FIG. **13**, when the timepiece **100** with the band **5** attached is to be worn on the wrist Wr of the user, the band **5** is wound around the wrist Wr, the pin **56** is locked in a small hole **58** corresponding to one's own wrist size, and the buckle **55** is passed through. The remaining end of the band main body **50** (the end on the side of the insertion-side end **50***b*) is passed through the free loop **57** and secured therein. When the band **5** inserted through the opening **35** is worn on the wrist Wr, both ends of the band **5** are pulled in the direction along the wrist Wr (indicated by an arrow in FIG. **13**). As a result, the second part **52**, which is relatively thin, flexes and deforms to fit the wrist Wr, and the extension part **34** in which the opening **35** is provided elastically deforms to flex inward (the side indicated by the arrow in FIG. **13**, in the direction toward the timepiece main body **1**), and the protruding part **36** of the opening **35** is pressed against the step part **53** when the extending part **34** is in the flexed state. This allows the timepiece **100** of this embodiment to be worn on a target object such as the wrist Wr, and the band **5** is firmly fixed to the bezel **3**, more reliably preventing it from falling off. In this state, the timepiece main body **1** is also sandwiched between the band **5** and the bezel **3** and is securely fixed in place.

[0062] When removing the bezel **3** from the timepiece main body **1**, first, the band **5** is removed from the opening **35** in the extending part **34** of the bezel **3**. When removing the band **5** from the bezel **3**, the band **5** is pulled out in the direction opposite to the insertion direction X after it has been removed from the wrist Wr. After the band **5** is removed, the extending parts **34** of the bezel **3** are pushed apart (outwardly in the directions of the 6 o'clock side and the 12 o'clock side). The first engaging part **37** as a member-side locking part provided on each of the extending parts **34** is thereby removed from the underside of the first protruding part **17** of the timepiece body **1**, and the engagement state is thereby released. The lateral parts **32** of the bezel **3** are also pushed apart in the directions in which they separate (outward in the direction of the **3** and 9 o'clock sides), and the operation units **14** inserted through the engagement holes **33** are removed. The fitted state is thereby released. This allows the band **5** and bezel **3** to be removed from the timepiece main body **1**.

[0063] As described above, the bezel **3** of the timepiece **100** of the present embodiment can be easily attached to and detached from the timepiece main body **1**, and it is possible to replace it with a different bezel **3** having a different color, material, shape, etc., or to replace the band **5** attached to the bezel **3**. Furthermore, it is also possible to attach a member other than the bezel **3** (such as an exterior case **7** shown in the second embodiment) to the timepiece main body **1**.

[0064] Next, the effects of the timepiece **100**, which is the electronic timepiece according to the first embodiment, and the timepiece main body **1** are described. The timepiece **100**, which is the electronic timepiece according to the present embodiment, includes the timepiece main body **1** and the bezel **3** as the exterior member that is attached to the timepiece main body **1** and can hold the band **5**. The bezel **3** has the opening **35** (first opening) on one end, and the opening **35** has the protruding part **36** (first protruding part) protruding inside the opening **35** on the side **35***a* facing the back side of the band **5** inserted through the opening **35**. The band **5** has the second part **52** (first planar part) on the back side and the first part **51** (second planar part) that swells more than the second part **52** on the back side. The boundary between the second part **52** and the first part **51** forms the step part **53**. When the band **5** is inserted through the opening **35** of the bezel **3**, the step part **53** and the protruding part **36** are engaged with each other, thereby holding the band **5** to the bezel **3** and the timepiece main body **1** with the bezel **3** attached.

[0065] As described above, in the present embodiment, as the band **5** is attached to the bezel **3** attached to the timepiece main body **1**, the timepiece main body **1** is sandwiched and fixed between the band **5** and the bezel **3**, and it is possible to ensure that the timepiece **100** will not easily come off when worn on the wrist Wr. On the other hand, the band **5** is only engaged with the protruding parts **36** of the bezel **3**, and can be easily removed by releasing the part hooked on the protruding parts **36** (the step parts **53** in the embodiment). Thus, the user can easily attach and detach the band **5** to and from the timepiece main body **1**, and can enjoy changing the band **5** and other parts.

[0066] In a conventional structure (for example, the structure disclosed in JP2005010905A), an engaging recess that engages with a locking protrusion on the inner peripheral surface of an exterior member (bezel) so that both parts are engaged with each other to attach the bezel to the case main body. However, in such a conventional structure, as the locking protrusion extends outward from the original outer shape of the case main body (see FIG. **2** of JP2005010905A), the outermost shape of the case main body is larger, resulting in an increase in the size of a timepiece. In this regard, in the structure shown in the embodiment, the external size of the timepiece main body to which an exterior member is attachable can be smaller.

[0067] In the embodiment, the bezel **3** (at least the protruding parts **36** of the bezel **3**) is elastically deformable, and when the band **5** is inserted to the opening **35**, the band **5** is inclined along the insertion direction X to pass through the opening **35** and is back to the original state at the second part **52** where it is thinner. The band **5** is thereby engaged with the step parts **53** which is the boundary with the first part **51**. Therefore, the band **5** can be pulled through the opening **35** smoothly with little resistance, and the elastic deformation of the protruding parts **36** can be used to ensure that the band **5** is securely engaged with the step parts **53**. This allows the user to easily put on and take off the band **5** without any stress.

[0068] Furthermore, as described above, the bezel **3** of the embodiment is detachable from the timepiece main body **1**, so that it can be easily replaced with another bezel **3** having a different color, material, shape, etc., or the band **5** attached to the bezel **3** can be replaced. Various exterior cases other than the bezel **3** can be easily attached.

[0069] The bezel **3** in the embodiment has the pair of the extending parts **34** that extend downward from the back face of the timepiece main body **1** at opposing positions across the timepiece main body **1** when the bezel **3** of the embodiment is attached to the timepiece main body **1**, and the pair of the openings **35** are respectively formed in the pair of the extending parts **34**. Thus, even when the timepiece main body **1** is to be used as a wristwatch with a band, it is not necessary to provide a band attachment portion on the timepiece main body **1**, and the timepiece main body **1** can be compact. Furthermore, as the band **5** is pulled through the pair of the openings **35**, the timepiece main body **1** can be held from the back side. It is thereby possible to more reliably prevent the timepiece main body **1** from coming off the bezel **3**.

[0070] In the embodiment, when the band **5** inserted through the opening **35** is worn on the wrist Wr, the extending parts **34** are pulled and bent toward the wrist Wr to be elastically deformed. The protruding parts **36** of the openings **35** are pressed against the step parts **53** in the state where the extending parts **34** are bent. This makes it difficult for the band **5** and the timepiece main body **1** to come off the bezel **3** in the state where the band **5** is worn on the wrist Wr because the timepiece main body **1**, the bezel **3**, and the band **5** are more securely integrated.

[0071] The timepiece main body **1** in the embodiment is provided with the first engaging parts **37** at the positions corresponding to the first protruding parts **17** when the bezel **3** is attached thereto. The bezel **3** is locked to the timepiece main body **1** as the first engaging parts **37** are engaged with the first protruding parts **17**. The timepiece main body **1** in the embodiment has the operation units **14** as the second protruding parts beside the first protruding parts **17**, and the bezel **3** as the first member has the engagement holes **33** at the positions corresponding to the operation units **14**. When the bezel **3** is attached to the timepiece main body **1**, the bezel **3** is locked to the timepiece main body **1** as the first engaging parts **37** are engaged with the first protruding parts **17** and the operation units **14** are fitted to the engagement holes **33**. Thus, the bezel **3** can be locked to the timepiece main body **1** with a stronger locking force than when the bezel **3** is locked to the timepiece main body **1** with the engagement between the first engaging parts **37** and the first protruding parts **17**.

[0072] The engagement holes **33** are through holes through which the operation units **14** are inserted, and the operation units **14** are exposed from the engagement holes **33** in the state where the bezel **3** is attached to the timepiece main body **1**. Thus, the user can operate the operation units **14** of the timepiece main body **1** while the bezel **3** is attached thereto. And as the operation units **14** are locked inside the through holes, it is possible to gain the locking force to lock the bezel **3** to the timepiece main body **1** just by fitting the operation units **14** to the engagement holes **33**.

[0073] Especially, the timepiece main body **1** in the present embodiment is formed in an approximately rectangular shape in plan view, and the first protruding part **17** is provided on each of two opposing sides of the timepiece main body **1** (the lateral faces on the 6 and 12 o'clock sides of the timepiece), and the second protruding parts, which are the operation units **14**, are provided on the lateral faces on the 3 and 9 o'clock sides of the timepiece, which are different from the sides on which the first protruding parts **17** are provided. Thus, when the first engaging parts **37** and the first protruding parts **17** are engaged with each other and the operation units **14** are fitted into the engagement holes **33**, the timepiece main body **1** can be locked on three or more sides, and the bezel **3** can be more stably locked to the timepiece main body **1**.

[0074] The back cover **12** is screwed to the back face side of the timepiece main body **1**, and the screws **15** that fix the back cover **12** are arranged on the path through which the band **5** passes when the band **5** is inserted to the opening **35** of the bezel **3**. The protective protrusions **18** that protrude beyond the screw heads are provided near the screws **15**. This can prevent the surface of the band **5** from interfering with the screw **15** and being damaged. In the embodiment, the protruding parts **36** are provided on the sides **35***a* of the openings **35** facing the back surface of the band **5** and protrude into the openings **35**, and are positioned so as to sandwich the band **5** from both the front and back with the protective protrusion **18** provided on the back face side of the timepiece main body **1**. Thus, the protective protrusions **18** can prevent the band **5** from shifting and falling off in combination with the engagement between the protruding parts **36** and the step parts **53**. The effect of holding the band **5** more securely can be expected thereby.

SECOND EMBODIMENT

[0075] In the second embodiment, a timepiece **200** includes the timepiece main body **1** and the exterior case **7**. The shape, configuration, etc. of the timepiece main body **1** are the same as those described above, so the description thereof is omitted. The exterior case **7** that constitutes the timepiece **200** is a case that is attached to the timepiece main body **1** so as to surround the outer periphery of the timepiece main body **1**, and is a second member that is selectively attached to the timepiece main body **1** in place of the bezel **3** that is the first member shown in the first embodiment.

[0076] As shown in FIG. **14** etc., the exterior case **7** of the present embodiment has a case main body **71** that is approximately circular in plan view and a locking part **73** that protrudes from the case main body **71**, and has a locking hole **731** formed on the outer side (the upper outer peripheral part in FIG. **14**, etc.) of the case main body **71**. For example, as shown in FIG. **18**, the locking hole **731** is a through hole that penetrates from front to back, and a ring, carabiner, strap, rope, etc. can be hooked or passed therethrough. The exterior case **7** is formed of silicone rubber or the like that has a relatively high rubber hardness, for example. In the embodiment, the exterior case **7** is made of a transparent or semi-transparent material, and the internal configuration can be seen from the outside. The material that forms the exterior case **7** is not limited to the materials illustrated herein.

[0077] The exterior case **7** according to the embodiment is open on both the front and back. As shown in FIGS. **14** and **15**, the opening **72** on the front side is formed to be approximately the same size as or slightly smaller than the size of the display panel **13** of the timepiece main body **1** so that when the exterior case **7** is attached to the timepiece main body **1**, at least part of the display panel **13** of the timepiece main body **1** is exposed, making various information such as the time visible. As shown in FIGS. **16**, **17**, etc., the opening **75** on the back side of the exterior case **7** is formed in the size large enough to put in and out the timepiece main body **1**. As shown in FIGS. **15**, **16**, etc., a back cover **74** is attached to the opening **75** on the back side and covers the opening part. When the back cover **74** is attached, part of the inner edge of the opening **75** is caught by the back cover **74**. The part caught by the back cover **74** is formed in a hooked shape, and stops the back cover **74** not to be detached. The back cover **74** is formed of a transparent acrylic plate, for example, and the contents inside can be seen from the outside.

[0078] A housing part **76** that houses the timepiece main body **1** is provided inside the exterior case **7**. As shown in FIG. **17**, etc., the housing part **76** is formed in the size and shape that substantially conform to the outermost shape Es (see FIG. **2**) of the timepiece main body **1**. As shown in FIG. **17**, for example, a corresponding inner wall of the housing part **76** is in contact with each of the sides on the 1 to 2, 4 to 5, 7 to 8, and 10 to 11 o'clock of the timepiece main body **1**. As a result, the timepiece main body **1** disposed inside the housing part **76** can be positioned and prevented from shifting and looseness. As shown in FIG. **17**, for example, on the 3 and 9 o'clock sides of the timepiece main body **1** where the operation units **14** are provided, sufficient space is provided to prevent the operation units **14** from coming into contact with the inner wall of the housing part **76** so that the operation units **14** are not pressed accidentally when subjected to vibrations, etc.

[0079] The exterior case **7** as the second member has second engaging parts **77** as a member-side locking part stopped by the first protruding part **17** when the exterior case **7** is attached to the timepiece main body **1**. In the embodiment, as shown in FIGS. **17** and **18**, the second engaging parts **77** as the member-side locking part are respectively provided at the positions corresponding to the first protruding parts **17** of the timepiece main body **1** in the housing part **76**. The second engaging parts **77** are a pair of tongue-shaped members that protrude inward from the inner wall of the housing part **76**, corresponding to the 6 and 12 o'clock positions of the timepiece main body **1** housed in the housing part **76**. In the state where the timepiece main body **1** is housed in the housing part **76**, the second engaging parts **77** are in contact with the first protruding parts **17** from the back side (the back cover **12** side of the timepiece main body **1**) and engage with the first protruding parts **17**. In the present embodiment, as the second engaging parts **77** and the first protruding parts **17** are engaged with each other, the exterior case **7** is locked to the timepiece main body **1**.

[0080] In the first embodiment, when the bezel **3** is attached to the timepiece main body **1**, the first engaging parts **37** are engaged with the first protruding parts **17** and the operation units **14** are fitted to the engagement holes **33** so that the bezel **3** is locked to the timepiece main body **1**. In the present embodiment, on the other hand, when the exterior case **7** is attached to the timepiece main body **1**, the exterior case **7** is locked to the timepiece main body **1** only by the engagement between the second engaging parts **77** and the first protruding parts **17**. Therefore, in order to fix the exterior case **7** to the timepiece main body **1**, the locking force of the second engaging parts **77** to the timepiece main body **1** is preferably larger than that of the first engaging parts **37** to the timepiece main body **1**. For this reason, the second engaging part **77** is thicker than the first engaging part **37**. Further, the exterior case **7** is formed of a relatively hard material such as silicone rubber, and its rigidity is higher than that of the bezel **3**. Thus, the timepiece main body **1** is sufficiently fixed only by the engagement between the second engaging parts **77** and the first protruding parts **17**.

[0081] In the timepiece main body **1** in the present embodiment, as described above, the adjacent ones of the screw hole spaces **16** are connected by the first protruding part **17** so that they are aligned substantially flush in plan view. Thus, the outermost shape Es of the timepiece main body **1** is in a shape comparatively straight without much unevenness, and when the exterior case **7** is formed of a transparent or semi-transparent member and the timepiece main body **1** is seen from the outside, the design can be of high quality and the appearance can be enjoyed.

[0082] The back face side of the exterior case **7** has a slight gap between the timepiece main body **1** and the back cover **74** that closes the opening **75** even in the state where the timepiece main body **1** is housed in the housing part **76**. In the present embodiment, as the back cover **74** is transparent, if a photograph or the like is inserted, it can be seen from the outside through the back cover **74**, and the back face side of the exterior case **7** can be customized according to the user's preferences. It is also possible that a logo or the like on the back cover **12** side of the timepiece main body **1** is to be enjoyed without a photo or the like inserted on the back face side of the exterior case **7**.

[0083] Next, the effects of the timepiece **200**, which is the electronic timepiece according to the second embodiment, are described. In the second embodiment, when the exterior case **7** as the second member is attached to the timepiece main body **1**, as shown in FIGS. **15**, **17**, etc., the timepiece main body **1** is housed in the housing part **76** from the opening **75** on the back face side of the exterior case **7**. At this time, the timepiece main body **1** is pressed into the housing part **76** so that the first protruding parts **17** provided on the 6 and 12 o'clock sides of the timepiece main body **1** go over the second engaging parts **77** as the member-side locking parts to be fitted to a predetermined position of the housing part **76**.

[0084] As a result, once the timepiece main body is housed at the predetermined position inside the housing part **76**, the pair of the second engaging parts **77** are in contact with the first protruding parts **17** from the back side (see FIG. **18**), the second engaging parts **77** are engaged with the first protruding parts **17** and the exterior case **7** is thereby locked to the timepiece main body **1**. Beside at the part where the operation units **14** are housed, the inner wall of the housing part **76** is in contact with the outer periphery of the timepiece main body **1**, and the timepiece main body **1** is thereby prevented from shifting and rattling. Thereafter, the opening **75** is closed by fitting the back cover **74** to the opening **75** on the back side of the exterior case **7**. In this way, the timepiece is complete (i.e. assembled). A strap, carabiner, or the like is attached to the locking part **73** in this state, the timepiece **200** can be attached to a bag or the like and carried around with the timepiece main body **1** therein.

[0085] Furthermore, when the exterior case **7** is detached from the timepiece main body **1**, first, the back cover **74** is detached from the opening **75** on the back face side of the exterior case **7**, and the top and bottom of the exterior case **7** where the second engaging parts **77** are provided (the upper and lower parts in FIG. **14**, the part corresponding to the 6 and 12 o'clock sides of the timepiece main body housed inside) are bent toward the front side of the exterior case **7**. At this time, the timepiece main body **1** can be pressed out from the front side to the back side of the exterior case. Once the second engaging parts **77** come off the first protruding parts **17** to release the engaged state as a result of this, the timepiece main body **1** can be detached from the exterior case **7**. The user can attach the bezel **3** as the first member to the timepiece main body so that it can be worn on the wrist Wr, or enjoy switching to the exterior case in a different color or shape.

[0086] Next, the effects of the timepiece **200**, which is the electronic timepiece according to the second embodiment and the timepiece maim body are described. In addition to the exterior case **7**, the bezel **3** according to the first embodiment or the like can be attached to and detached from the timepiece **200** in the present embodiment. The bezel **3** and the exterior case **7** can be selectively attached to the timepiece main body **1**. Thus, just by changing the attachment, one timepiece main body **1** can be used as a wristwatch worn on the wrist Wr, as an ornament with a strap, and in other various forms.

[0087] As described hereinbefore, the timepiece main body **1** has the screw hole spaces **16** as protrusions which protrude outward and the apexes of which constitute the outermost shape Es (see Fig. Es), and the adjacent ones of the screw hole spaces **16** are connected by the first protruding part **17** so that they are aligned substantially flush in plan view. Therefore, the outermost shape Es of the timepiece main body **1** is in a shape comparatively straight without much unevenness, and when the exterior case **7** is formed of a transparent or semi-transparent member and the timepiece main body **1** is seen from the outside, the design can be of high quality and the appearance can be enjoyed.

[0088] Furthermore, on the timepiece main body **1** in the present embodiment, the first protruding parts **17**, which stop the member-side locking parts (the first engaging part **37**, the second engaging part **77**) when the bezel **3** as the first member or the exterior case **7** as the second member is attached to the timepiece main body **1**, are formed not to protrude outward beyond the apexes of the connected screw spaces **16** on the protruding side. Therefore, the structure for attaching the bezel **3** or the exterior case **7** to the timepiece main body **1** can be realized without changing the outermost shape Es of the timepiece main body **1** (without increasing the size of the timepiece main body **1**).

[0089] The exterior case **7** as the second member in the present embodiment has the housing part **76** that houses the timepiece main body **1** therein, and the second engaging parts **77** as the member-side locking part are provided at the positions corresponding to the first protruding parts **17** in the housing part **76**. Therefore, in the case where the exterior case **7** is attached to the timepiece main body **1**, the timepiece main body **1** is easily positioned just by being housed inside the housing part **76**, and the first protruding parts **17** are locked to the second engaging parts **77** too.

[0090] Although an embodiment of the present disclosure is described above, needless to say the present disclosure is not limited to the embodiment and can be appropriately modified in a variety of aspects without departing from the scope of the present disclosure. For example, in the present embodiments, the timepiece main body **1** is substantially rectangular in plan view, but the timepiece main body **1** that constitutes the timepiece **100**, **200**, which is the electronic timepiece in the present disclosure is not limited to a rectangular shaped one. For example, it may be substantially circular or substantially elliptical in plan view. In such cases, the first protruding parts are preferably formed at positions facing each other on the timepiece main body **1**. This makes it possible to lock the timepiece main body **1** to the first member or the second member in a good balance at least at the position of the first protruding part, making it less likely for the timepiece main body **1** to come off.

[0091] Further, in the present embodiment, for example, the engagement holes **33** provided in the bezel **3** as the first member are through holes through which the operation units **14** as the second protruding parts are inserted, but the engagement holes are not limited to through holes. For example, they may be notched parts that can be hooked onto part of the operation units **14**. In such a case, the locking force with the through holes cannot be expected, but if the notched parts are shaped to cover the operation units **14** from the front side (the front face side of the timepiece, the upper side in FIG. **1**), for example, the operation units **14** can be locked from the front side and the first protruding parts **17** are locked by the first engaging parts **37** from the back side so that the timepiece main body **1** is pressed from the front and back. The effect of preventing looseness and shifting can be thereby expected. Compared to the case where the engagement holes **33** are through holes, in the case where they are notched parts, the bezel **3** can be easily attached to and detached from the timepiece main body **1**.

[0092] The timepiece main body **1** may not have the second protruding parts that protrude on the 3 and 9 o'clock sides. That is, the timepiece main body **1** may not have operation units **14** that protrude outward. In such a case, the bezel **3** does not need to have the engagement holes **33**. The bezel **3** is locked to the timepiece main body **1** just by the engagement between the first protruding parts **17** and the first engaging parts **37**. However, even in that case, the band **5** is positioned on the back face side of the timepiece main body **1** when the band **5** is attached to the timepiece main body **1**, and the timepiece main body **1** is sandwiched between the bezel **3** and the band **5** from the front and back and is prevented from accidentally coming off.

[0093] For example, in the present embodiment, the timepiece main body **1** is housed in the housing part **76** of the exterior case **7** as the second member in the state where the 12 o'clock side faces up. However, in the case where the shapes of the four sides of the timepiece main body **1** are approximately the same as in the present embodiment, the timepiece main body **1** may be housed in a different orientation (for example, upside down or sideways) in the housing part **76**. For example, when the timepiece main body **1** is used in the exterior case **7**, it is housed in the housing part upside down (with the 12 o'clock side facing downwards) so that its display is lifted to the reverse orientation to be viewed like a fob watch. When the timepiece main body **1** is housed sideways, the second engaging part **77** is positioned between two of the operation units **14** on the same side. As a space equal to the space between the protective protrusions **18** for the first protruding part **17** is provided between those two operation units **14**, the second engaging part **77** can be locked between the operation units **14** while avoiding the operation units **14**, in the present embodiment.

[0094] In the present embodiments, although the exterior case **7** as the second member is in a substantially circular shape in plan view, for example, the shape or the like of the exterior case **7** is not limited to the illustrated example, and may be approximately elliptical or approximately rectangular in plan view. Furthermore, the exterior case **7** may be shaped like a character, an animal, or the like.

[0095] The timepiece main body **1** constituting the electronic timepiece **100**, **200** according to the present disclosure includes a wide range of devices having a function of time display, and can be applied to an electronic device such as a smart watch and sports watch, a wearable device that acquires biological information such as heartbeat and blood flow in addition to giving time information.

[0096] In the above, one or more embodiments of the present disclosure are described. However, the scope of the present disclosure is not limited thereto, and includes the scope of claims below and the scope of their equivalents.

What is claimed is:

1. An electronic timepiece comprising:

a timepiece main body; and

an exterior member attachable to the timepiece main body,

wherein the timepiece main body includes:

a plurality of protrusions which protrude outward on an outer periphery of the timepiece main body and whose apexes on extending sides form an outermost shape of the timepiece main body; and

a first protruding part that is formed to connect adjacent ones of the plurality of protrusions and do not protrude outward more than apexes on extending sides of the connected ones of the plurality of protrusions,

wherein the exterior member has an engaging part that is engaged with the first protruding part when the exterior member is attached to the timepiece main body.

2. The electronic timepiece according to claim **1**,

wherein the exterior member is a first member or a second member selectively attached to the timepiece main body,

wherein the first member has a first engaging part at such a position that the first engaging part is engaged with the first protruding part when the first member is attached to the timepiece main body,

wherein the second member has a second engaging part at such a position that the second engaging part is engaged with the first protruding part when the second member is attached to the timepiece main body.

3. The electronic timepiece according to claim **2**,

wherein the timepiece main body has a second protruding part,

wherein the first member has a first engagement hole at such a position that the first engagement hole is engaged with the second protruding part when the first member is attached to the timepiece main body,

wherein the first member is attached to the timepiece main body with the first engaging part being engaged with the first protruding part and the first engagement hole being fitted to the second protruding part, the first member thereby being locked to the timepiece main body,

wherein the second member is attached to the timepiece main body with the second engaging part being fitted to the first protruding part, the second member thereby being locked to the timepiece main body.

4. The electronic timepiece according to claim **3**,

wherein the first engagement hole is a through hole through which the second protruding part is inserted, and the second protruding part is exposed from the first engagement hole when the first member is attached to the timepiece main body.

5. The electronic timepiece according to claim **2**,

wherein a locking force of the second member to the timepiece main body with engagement between the second engaging part and the first protruding part is stronger than a locking force to the timepiece main body with engagement between the first engaging part and the first protruding part.

6. The electronic timepiece according to claim **3**,

wherein a locking force of the second member to the timepiece main body with engagement between the second engaging part and the first protruding part is stronger than a locking force to the timepiece main body with engagement between the first engaging part and the first protruding part.

7. The electronic timepiece according to claim **2**,

wherein the second engaging part is thicker than the first engaging part.

8. The electronic timepiece according to claim **3**,

wherein the second engaging part is thicker than the first engaging part.

9. The electronic timepiece according to claim **2**,

wherein the first member and the second member are respectively formed of resin materials,

wherein the resin material of the second member has a higher rigidity than the resin material of the first member.

10. The electronic timepiece according to claim **3**,

wherein the first member and the second member are respectively formed of resin materials,

wherein the resin material of the second member has a higher rigidity than the resin material of the first member.

11. The electronic timepiece according to claim **3**,

wherein the timepiece main body is formed in a substantially rectangular shape in plan view,

wherein the first protruding part is provided on each of two facing sides of the timepiece main body,

wherein the second protruding part is provided on a side different from the sides on which the first protruding part is provided.

12. The electronic timepiece according to claim **1**,

wherein the exterior member is a member transparent or semi-transparent,

wherein the first protruding part is formed such that adjacent ones of the protrusions are connected by the first protruding part to be aligned substantially flush in plan view.

13. The electronic timepiece according to claim **1**,

wherein the timepiece main body is formed in a substantially rectangular shape in plan view,

wherein the first protruding part is formed such that adjacent ones of the plurality of protrusions on a same side of the rectangular shape are connected by the first protruding part.

14. A electronic timepiece main body comprising:

a plurality of protrusions whose apexes on extending sides form an outermost shape of the timepiece main body; and

a first protruding part that is formed to connect adjacent ones of the plurality of protrusions and do not protrude outward more than apexes on extending sides of the connected ones of the plurality of protrusions,

wherein the first protruding part is engaged with an engaging part of an exterior member attachable to the timepiece main body.

15. The electronic timepiece main body according to claim **14**,

wherein the first protruding part is formed such that adjacent ones of the plurality of protrusions are connected by the first protruding part to be aligned substantially flush.

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